Request for Information: Development of an Artificial Intelligence Action Plan

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I. Hardware and Chips:

- **Domestic Manufacturing:** Incentivize semiconductor production in the U.S. through tax credits and public-private partnerships. This will reduce the dependency on Taiwan semiconductor manufacturing.
- **Supply Chain Resilience:** Establish strategic reserves of critical materials and foster partnerships with trusted allies to secure supply chains.

II. Data Centers:

- **Infrastructure Investment:** Support development of energy-efficient, high-capacity data centers through grants and streamlined permitting processes.
- **Resilience Standards:** Implement cybersecurity and physical security standards to protect critical AI infrastructure.

III. Energy Consumption and Efficiency:

- **Nuclear Energy Integration:** Promote the use of small modular nuclear reactors (SMRs) and other forms of nuclear power to provide reliable, low-carbon power for AI infrastructure.
- **Efficiency Standards:** Develop guidelines for sustainable AI development to optimize energy usage and reduce waste.

IV. Model Development:

- **Innovation Support:** Fund R&D for advanced AI architectures, promoting American leadership in foundational models.
- **Human Oversight:** Mandate human-in-the-loop systems for high-stakes AI applications, ensuring accountability.

V. Open-Source Development:

- **Secure Open-Source Ecosystem:** Invest in secure, verified open-source platforms while balancing IP protections to spur innovation.

VI. Application and Use:

- **Sector-Specific Guidelines:** Develop tailored guidelines for AI use in healthcare, finance, defense, and education.
- **Government Adoption:** Modernize federal agencies with AI capabilities, prioritizing systems that enhance public service.

VII. Explainability and Assurance:

- **Transparency Standards:** Require AI systems to provide human-readable explanations for critical decisions. Develop guidelines for standardized reporting that outlines model decision pathways, key influencing factors, and confidence levels. Encourage the use of visual and interactive tools to help users understand complex model behavior, especially in high-impact domains like healthcare, law, and public safety. Foster research into new techniques that improve interpretability without sacrificing model performance.
- **Verification Processes:** Establish independent bodies to certify AI system accuracy and reliability.

VIII. Cybersecurity:

- **Robust Defenses:** Mandate regular penetration testing and adversarial resilience for AI systems.
- **Threat Intelligence Sharing:** Foster public-private partnerships for real-time threat data exchange.

IX. Data Privacy and Security:

- **Lifecycle Protections:** Enforce data security practices across AI system lifecycles, including encryption and access controls.
- **AI Model Integrity:** Develop countermeasures against model poisoning and adversarial attacks.
- **Protections for Certain Types of Data:** Ensure protections for data including but not limited to personal identifiable information, sensitive, medical information, patient privacy, and confidential information.

X. Risks, Regulation, and Governance:

- **Risks:** AI can make mistakes (hallucinations), which is why it is important to have a human-in-the loop to verify and check AI outputs. Businesses can be encouraged to adhere to a confidence scoring system to filter hallucinations, which will create more reliable AI systems. The scoring system would also allow end users to know that they are interacting with a specific AI tool with a certain level of confidence.

- **Pornography Legislation:** Enact federal laws to address AI deepfake pornography including the offenses of possession, distribution, and manufacturing. Protection should apply to adults and children. Many states have already enacted that child pornography is illegal even if it is AI generated.
- **Limitations of AI Self-Awareness/Consciousness Regulations –** Collaborate with industry and academia to set evolving safety and performance benchmarks. Ensure that businesses put controls in place and create legislation to define and put limitations around AI from becoming self-aware or conscious. This would put the public's fears and skepticism at ease because movies such as the "Terminator" series and "2001: A Space Odyssey" are still very much at the forefront of American pop culture.
- **Proportionate Regulation:** Develop a flexible, risk-based regulatory framework that adapts to technological advances. High-risk applications, like healthcare and defense, should face stricter oversight, while low-risk, consumer-facing applications should experience lighter touch regulation. This approach encourages innovation while safeguarding public welfare. Regularly review and update regulations to keep pace with AI development, minimizing compliance burdens for smaller firms and startups.
- **Ethical AI Boards:** Create advisory panels with diverse stakeholders to guide AI governance. AI should also be addressed in the ethics of professional industries (state bar professional rules of responsibility, medical boards, etc.)
- **Unauthorized Practice of Certain Professions:** A major risk of AI for the public is for them to rely on AI outputs for legal, psychiatric, and medical care among other areas. Controls should be put in place to prevent this from happening or businesses should provide prominent disclaimers for consumers to not rely on the information regarding these areas.

XI. Technical and Safety Standards:

- **Standards Development:** Collaborate with industry and academia to set evolving safety and performance benchmarks.
- **Medical Industry:**
    - **Human-in-the-Loop:** Doctors should not rely solely on AI recommendations alone but should use AI technologies to assist them with patient diagnoses and treatment (human-in-the-loop verification).
    - **Standardizing Information:** Standardize the format of medical information to allow for consistency and inputs for medical AI technologies. Medical records need to be securely stored electronically and existing software systems should have the ability to extract data, which can then be converted into a standard

format to be fed into AI systems. However, this does not mean that there needs to be a national medical software system.

- **Minimize the Impact of AI Medical Technologies:** Minimize the impact on medical offices by using AI technologies, so they will not be frequently having to adapt and migrate to new systems. A current problem is that many medical software systems are unable to talk to each other (ex. primary care and specialist systems). Going to a new system is expensive with the cost of purchasing another software. Moreover, it takes time to train healthcare workers to learn another system. Instead, encourage research on finding ways to extract existing medical information from multiple systems and upload it into an AI model that can be made accessible to medical professionals in diagnosing and treating patients based on a current patient's test results and medical history.
- **Affordability and Accessibility:** Medical AI systems used for diagnostics and treatment should be affordable and accessible to the medical community. "Affordable" is broad and relative term, and businesses should make a profit. However, a doctor on an American Indian reservation should be able to access AI technologies the same as a doctor in a large hospital to help them the diagnosis and treatment of their patients. Federal, state, and local government entities as well as American Indian reservations should be eligible for AI technology discounts and trainings to level the playing field.

XII. National Security and Defense:

- **AI for Defense:** Expand AI adoption for threat detection, logistics, and strategic decision-making.
- **Guardrails for Dual-Use Tech:** Regulate AI applications with potential military and civilian uses.
- **National Guard:** Expand the role of the National Guard cybersecurity branch to provide additional support for AI cybersecurity by working with federal, state agencies, and local government agencies.

XIII. Research and Development:

- **R&D Funding:** Increase federal AI research budgets, prioritizing transformative technologies.
- **Innovation Hubs:** Establish AI research centers to bridge academia, industry, and government.

XIV. Education and Workforce:

- **AI Education Pipeline:** Invest in STEM programs, AI-focused curricula, and vocational training. Incentivize students to major in STEM fields in colleges and universities by creating a debt forgiveness program. Create grants or provide incentives for states who encourage and fund STEM curriculum grades K-12 in school systems. Encourage states to allow high school guidance counselors to play a more active role in evaluating and identifying students who would be ideal candidates to go into STEM fields. Create a national awareness program to encourage women to go into STEM fields.
- **Reskilling Initiatives:** Support displaced workers with AI-centric reskilling programs.

XV. Innovation and Competition:

- **Competitive Markets:** Prevent monopolistic practices and consumer price gauging, ensuring startups can compete and thrive.
- **Venture Capital Support:** Facilitate funding for AI startups through public and private channels.

XVI. Intellectual Property:

- **IP Protections:** Strengthen AI-related patent protections to safeguard American innovations. Establish clear guidelines for AI-generated inventions, ensuring that intellectual property laws keep pace with technological advancements. Enhance enforcement mechanisms to combat IP theft, particularly from foreign adversaries, by increasing penalties and investigative resources. Support international cooperation on AI patents to facilitate fair and reciprocal protections for U.S. businesses operating globally.
- **US Patent and Trademark Office Employees:** Hire more patent agents, specifically those with computer science educational backgrounds or STEM backgrounds to process AI patents quicker.
- **Fast-Track AI Patents:** Accelerate patent reviews for critical AI technologies in hardware, software, or other.
- **Pharmaceutical Industry:**
    - Implement a process allowing companies to share AI IP with other companies akin to the pharmaceutical industry where proven and tested drugs can be made into generics, so that companies entering the AI market do not have to start from nothing.
    - AI will likely speed up innovation in many areas and especially in the pharmaceutical industry. Update the Biologics Price Competition and Innovation Act (BPCIA) to allow FDA sharing of original biologics manufacturing information with follow-on biologics developers. This will create more competition for companies entering the biologics market because AI will allow

for faster and newer drug development. Moreover, by creating more competition in this area, there will be a biologics price reduction.

XVII. Procurement:

- **AI-Ready Procurement:** Modernize federal procurement practices to streamline AI system adoption.

XVIII. International Collaboration and Export Controls:

- **Strategic Alliances:** Build international AI research alliances while safeguarding U.S. interests. Specifically, the U.S. should partner with Israel.
- **Export Safeguards:** Tighten export controls on sensitive AI technologies to prevent adversarial misuse. Implement a tiered control system that classifies AI technologies based on their potential impact on national security. Enhance end-use monitoring and enforce strict penalties for violations. Collaborate with allies to establish common export standards, ensuring that critical technologies do not fall into the hands of adversaries while preserving the ability of U.S. companies to compete globally.

XIX. AI Litigation and the Judicial Branch:

- **Litigation:** AI related lawsuits will ensue. However, plaintiffs should be required to show standing. This will discourage repeat plaintiffs and prevent the systematic targeting of businesses with AI technologies. Encourage state and local governments to require plaintiffs to show standing who have been harmed by AI.

- **Judges:** Ensure that federal judges receive mandatory CLE training, specifically on AI. Encourage state and local judges to receive mandatory AI CLE training. Hire and appoint federal judges with STEM backgrounds.

Conclusion: This action plan aims to solidify America's leadership in AI, fostering innovation while protecting national interests and vulnerable populations. Human oversight remains a cornerstone, ensuring AI systems serve the public good without compromising safety or values. By balancing innovation with responsible stewardship, the United States can shape an AI-powered future that upholds American ideals and global leadership.